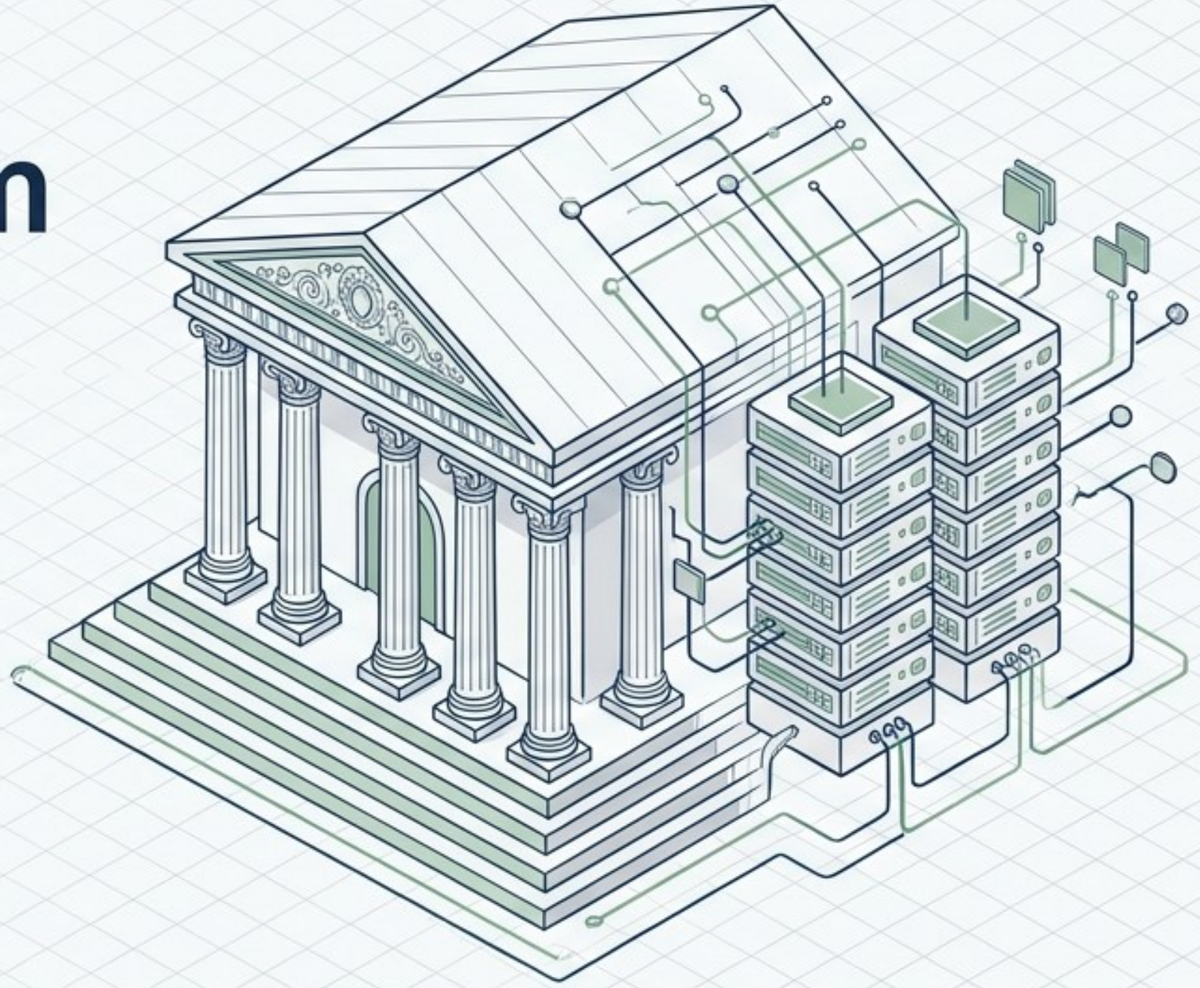
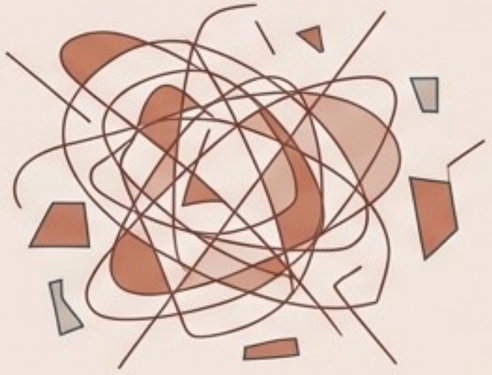
SQL Banking System

From Database Modeling to Strategic Business Analytics

A complete journey from systemic crisis to automated decision-making.

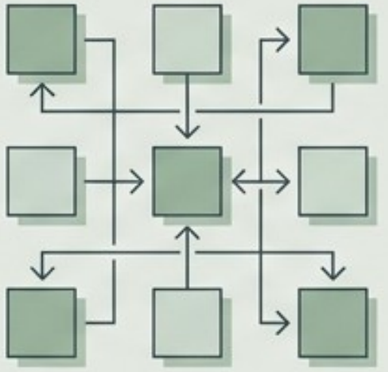


Scattered Data Replaced by Centralized Infrastructure



Problem: Systemic Crisis

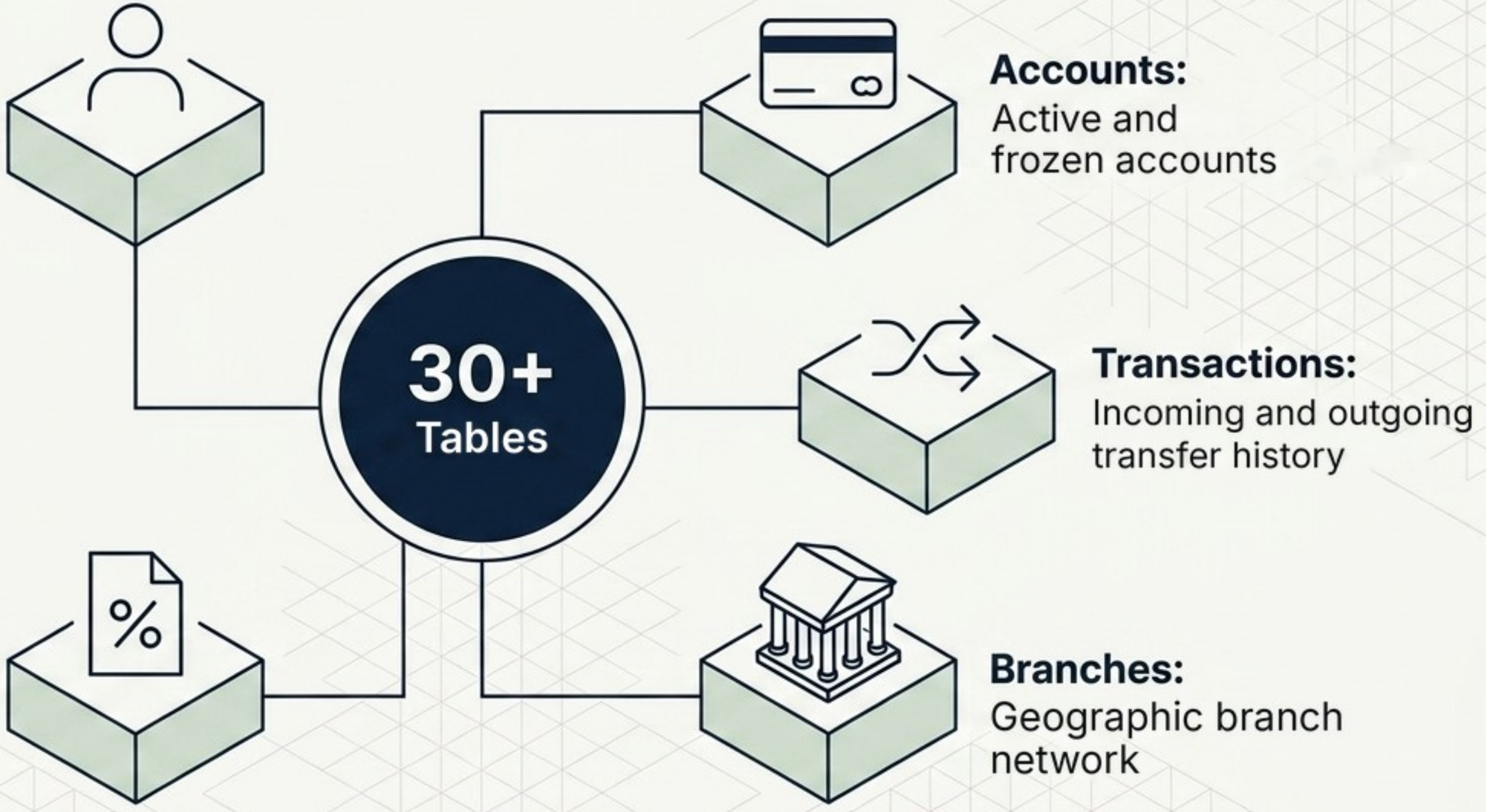
- Data fragmentation and disconnection
- Delays in generating reports
- Lack of transparency in decision-making



Goal: BankingSystem

- A single, centralized database powered by SQL Server
- High accuracy that adapts to business needs
- Solving data management problems at the root

A Centralized Architecture for All Operations



Customers:
All customer profiles

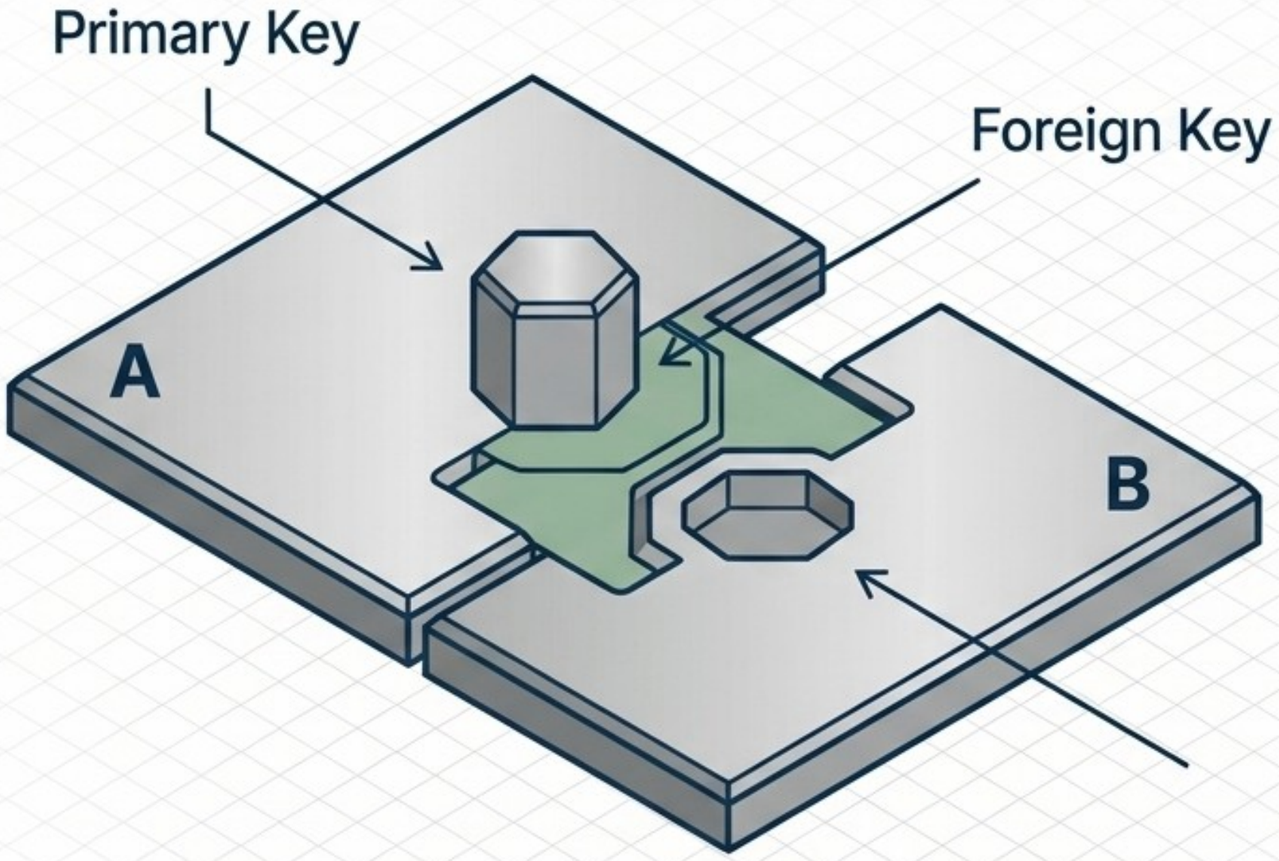
Accounts:
Active and frozen accounts

Transactions:
Incoming and outgoing transfer history

Loans:
Dynamic, ever-changing loan portfolio

Branches:
Geographic branch network

Relational Design: Error-Free System Guarantee



Mechanism

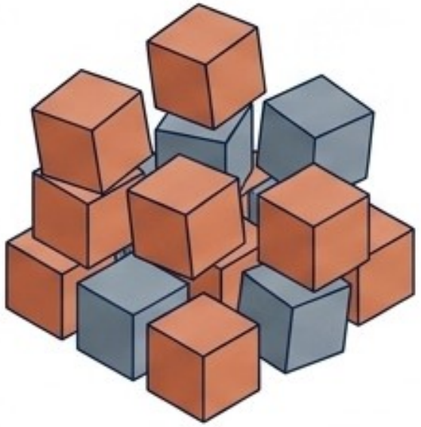
- **PK (Primary Key):** A unique numeric identifier for every record (the unrepeatable core).
- **FK (Foreign Key):** A strict logical and physical link between tables.

Business Value

- Data Integrity is 100% guaranteed.
- Systemic errors and orphan anomalies are blocked automatically.
- Separate tables operate in sync as a single, living organism.

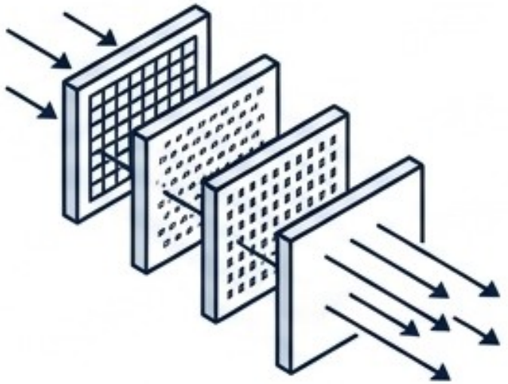
Data Quality: Cleaning 10,000+ Records

Problem (Messy Data)



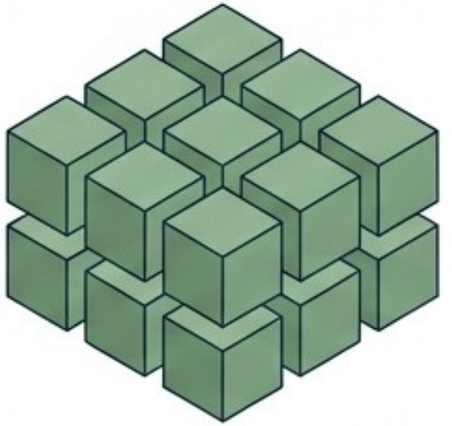
- NULL (empty) values
- Incorrect dates and formats
- Duplicate rows
- Broken ID records

Action (Data Cleaning)



- Strict validation checks
- Removing duplicates
- Standardizing formats
- Verifying ID consistency

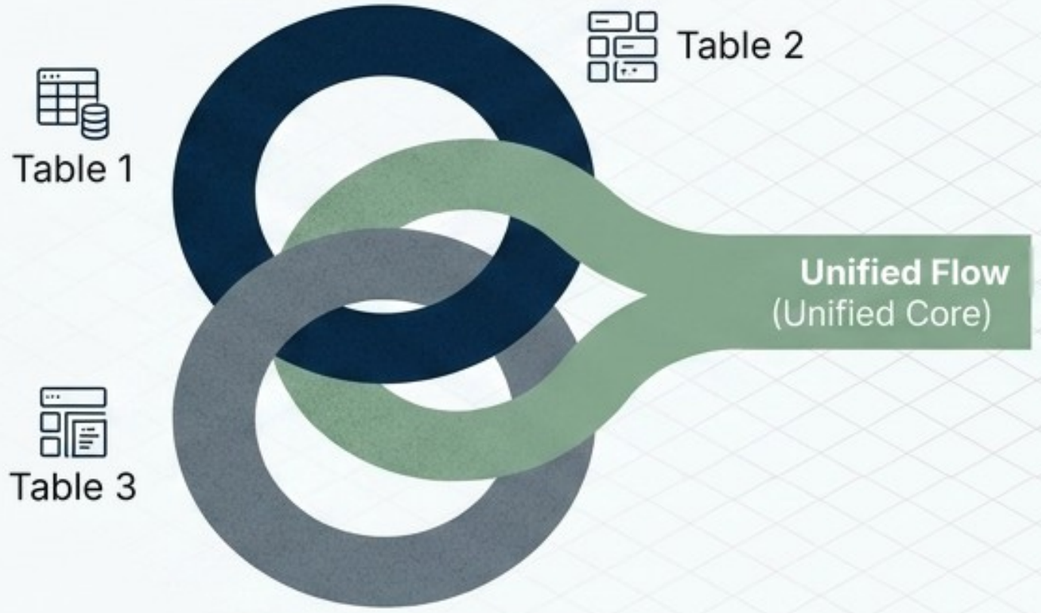
Result (Structured Database)



- 100% ready, clean, and reliable
- A gold-standard reserve of 10,000+ rows
- Ready for Big Data analytics

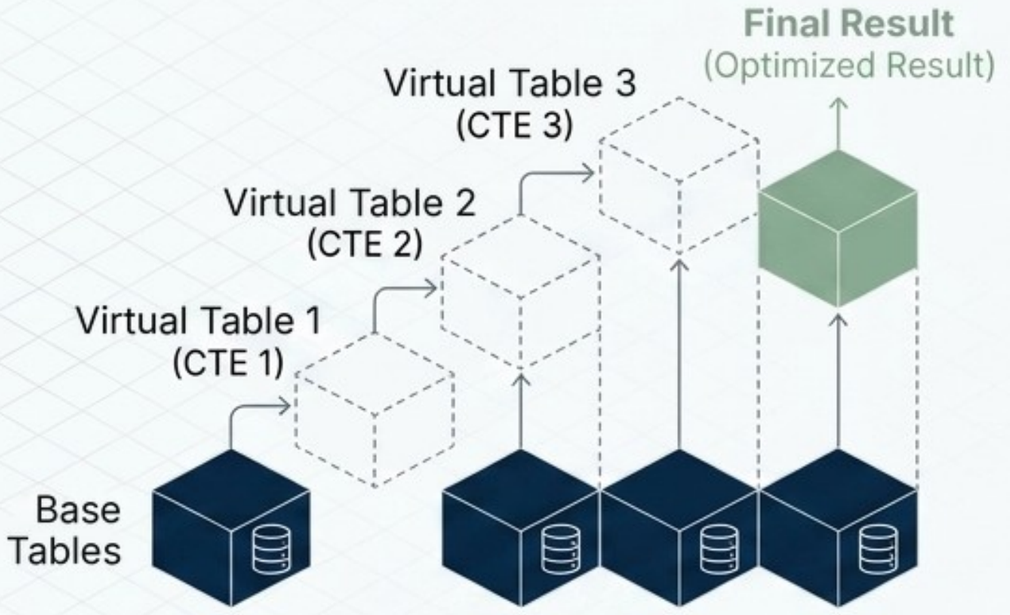
Analytics Engine: JOIN and CTE Technologies

JOIN (Merging Mechanism)



- Function: Gathering various scattered tables into one common stream.
- Result: A complete 360-degree view of all of the customer's interactions with the bank.

CTE (Common Table Expressions)

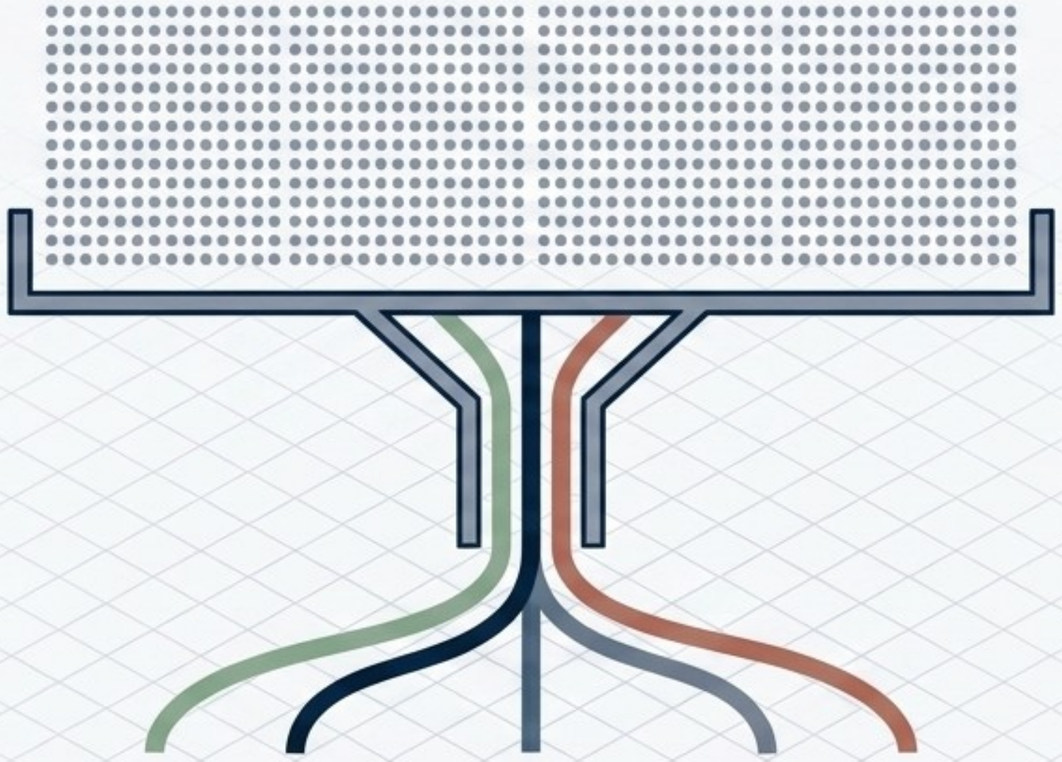


- Function: Creating a set of temporary results (virtual tables).
- Result: Simplifying complex, multi-step calculations and optimizing system memory.

Compressing Big Data: Aggregate Functions

Mechanism

- SUM (Total)
- COUNT (Number)
- AVG (Average)
- MAX & MIN



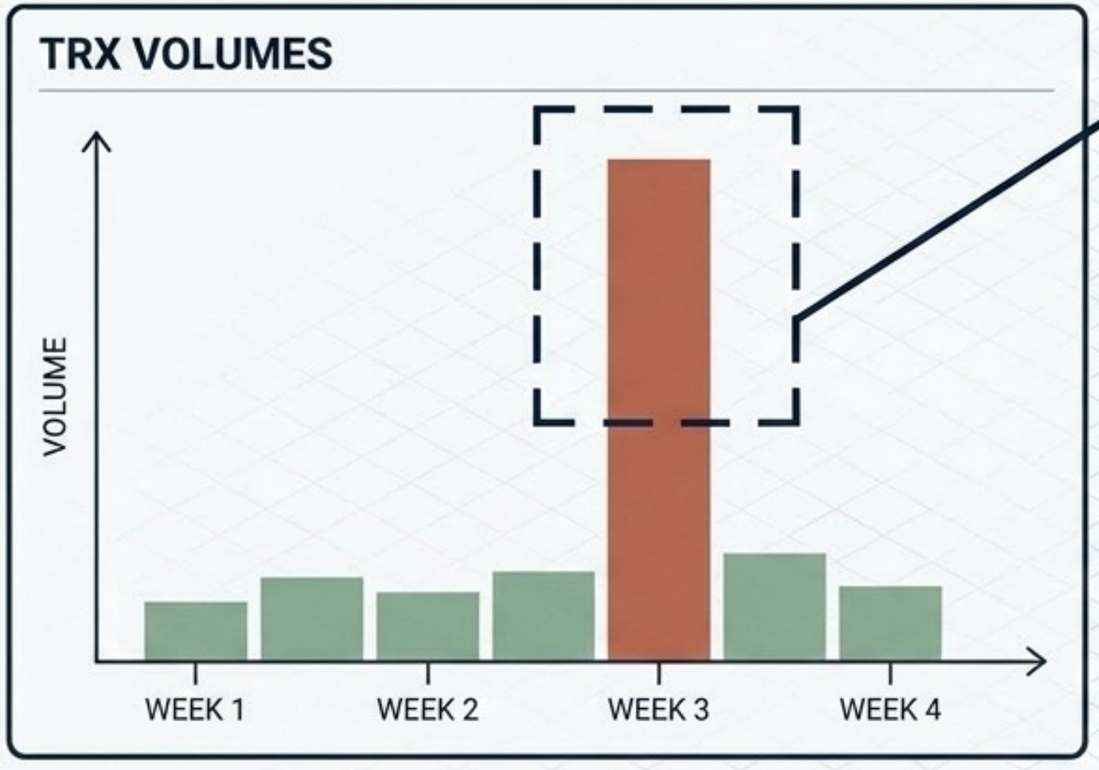
Process

- Grouping thousands of rows
- Calculating summary metrics in one second

Business Value

Turning financial trends, risk patterns, and strategic figures hidden in millions of rows into concise insights that are easy to act on.

Financial Security: Fraud Analysis



Critical Indicator (KPI): $10,000+ Transactions in a Short Time Window

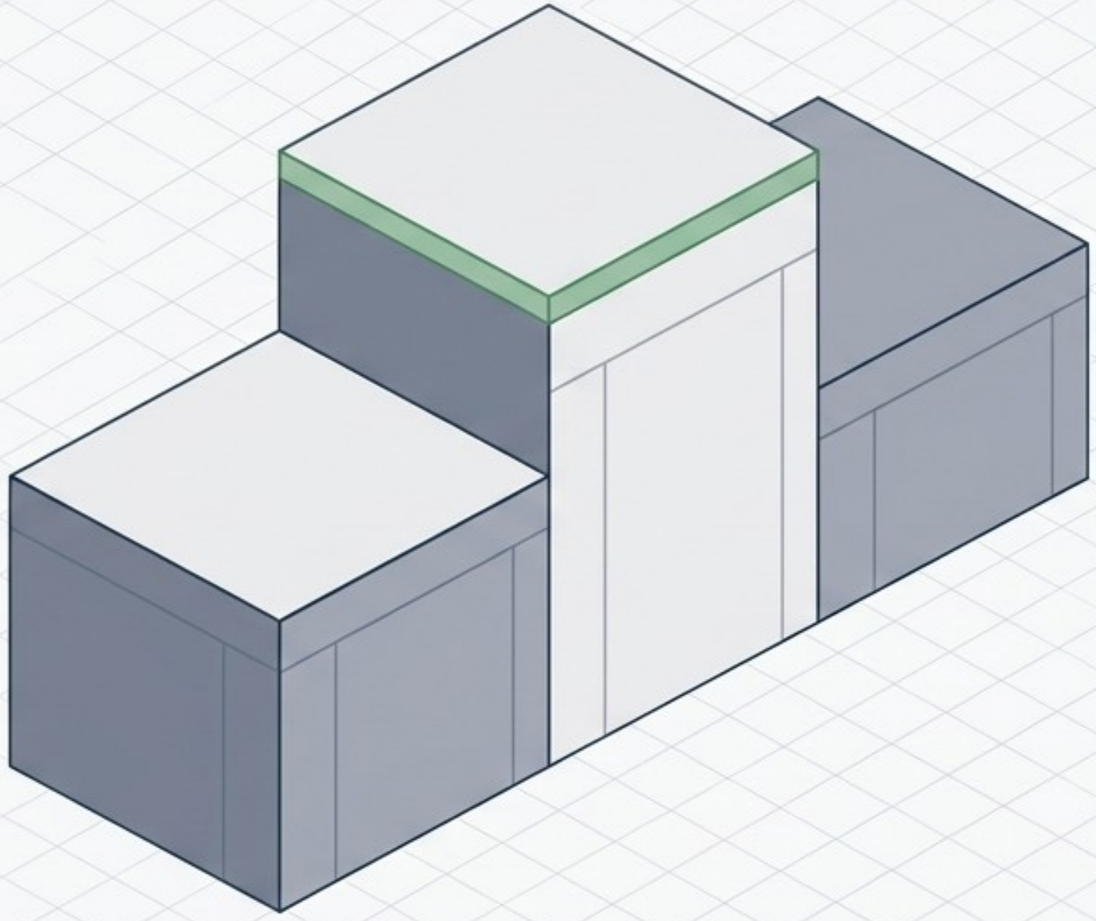
Analysis Logic

- Filtering transaction speed and volume in real time via SQL scripts.
- Instantly isolating abnormal activity and behavior.

Result (Risk Monitoring)

- A proactive system ensuring the bank’s financial security.
- Automatically blocking risks before they arise.

Increasing Revenue: VIP Segmentation



Goal

Identifying the VIP customer segment (Top-3) holding the most funds in the bank's overall portfolio.

SQL Solution

Building a dynamic list using Sorting and complex Ranking functions.

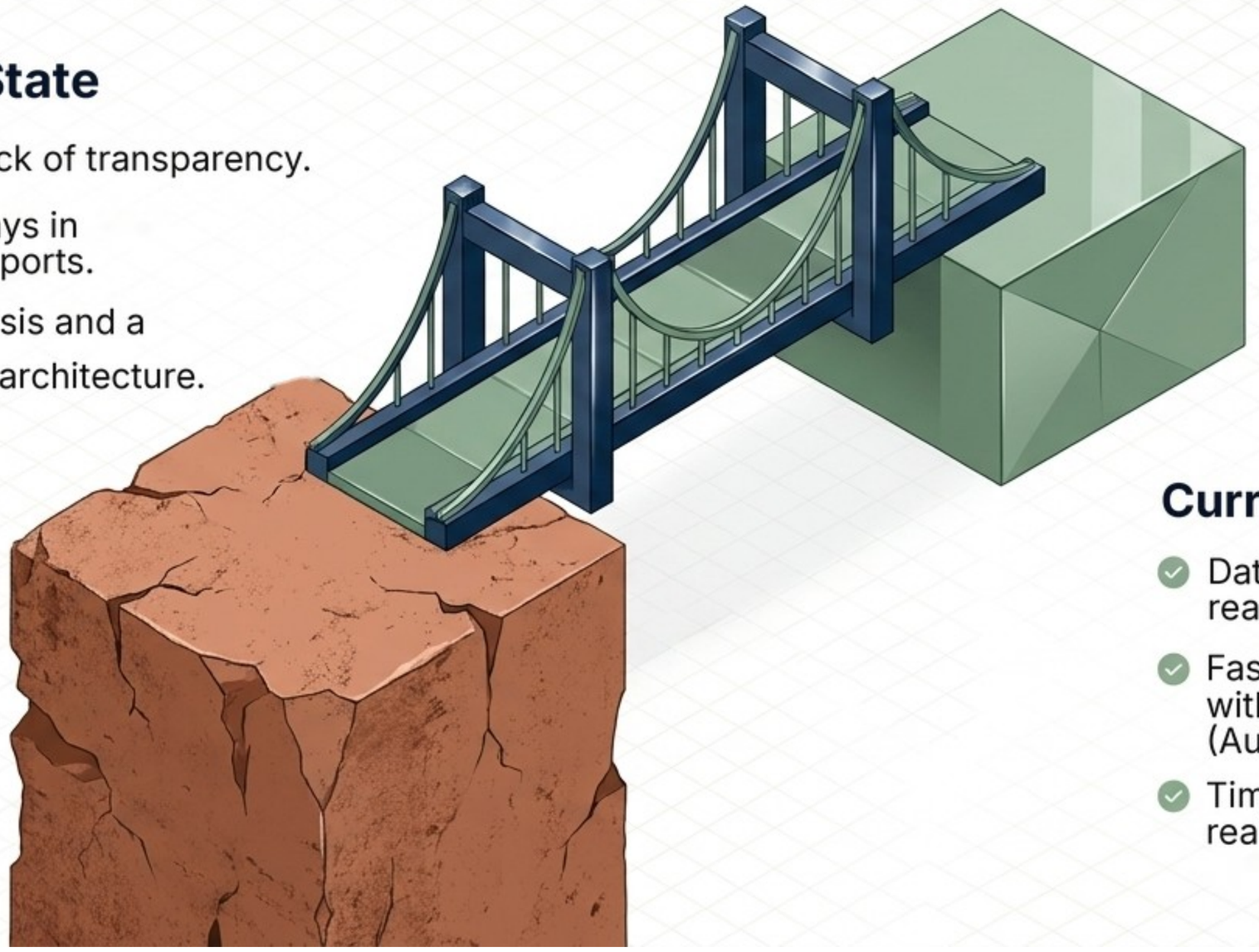
Business Value

- Launching exclusive customer Loyalty Programs.
- Protecting the bank's capital by offering personalized service to premium clients.

Project Outcome: Strategic Transformation

Previous State

- Complete lack of transparency.
- Delays of days in preparing reports.
- Systemic crisis and a fragmented architecture.



Current State (Result)

- Data integrity and accuracy reached 100%.
- Fast, automated decisions without the human factor (Automated Decision Making).
- Timely risk detection and real-time monitoring.

Development Roadmap: Next Steps



1. Power BI Integration

In-depth data visualization and building Real-time Dashboards for business owners.

2. Artificial Intelligence (AI)

Automatically assessing customer creditworthiness (Credit Scoring) and predicting risks.

3. Big Data Scaling

Adapting the system, via indexing methods, to process millions of transactions within seconds.

THANK YOU!

for your attention

Yunusov Javohir
Group F67

Banking System **SQL**